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Hunters Hill
High School

2020 Trial Examination

Mathematics Extension 2

**General
Instructions**

Reading time – 5 minutes
Working time – 3 hours
Write using black pen
Calculators approved by NESA may be used
A reference sheet is provided at the back of this paper
In Questions 11-16, show relevant mathematical reasoning and/or calculations

**Total Marks:
100**

Section I – 10 marks (pages 3-7)
Attempt all Questions 1-10
Allow about 15 minutes for this section

Section II – 90 marks (pages 8-13)
Attempt Questions 11-16
Allow about 2 hours and 45 minutes for this section

Section I

10 marks

Attempt Questions 1–10

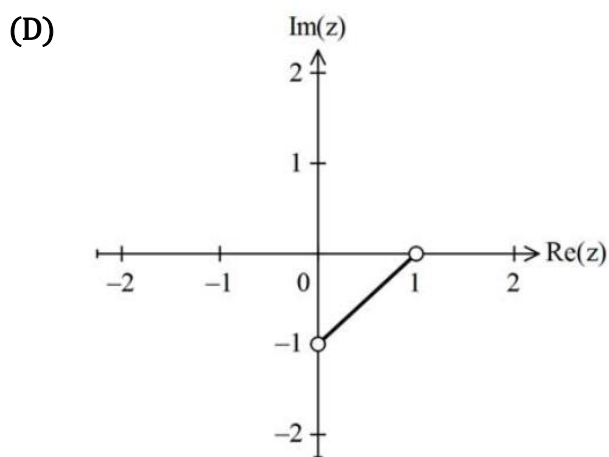
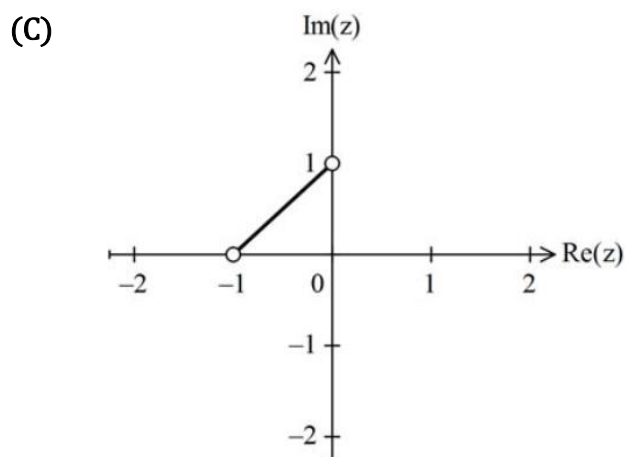
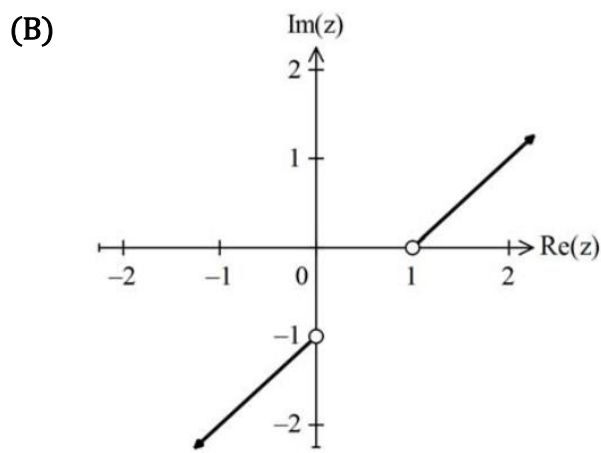
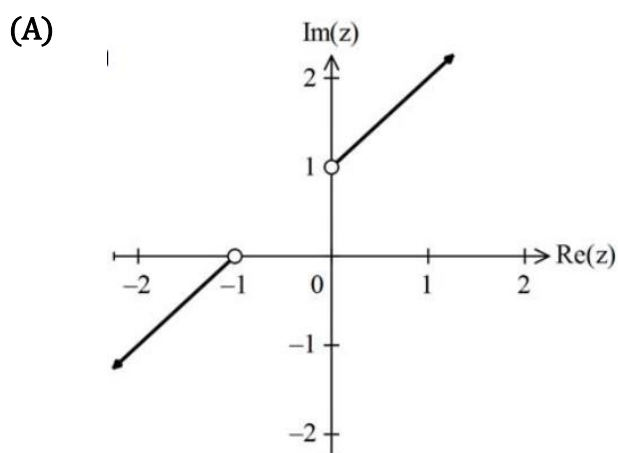
Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Question 1–10.

1. Imagine ω is an imaginary cube root of unity, then $(1 + \omega - \omega^2)^{2020}$ is equal to

- (A) $-2^{2020}\omega$
- (B) $2^{2020}\omega$
- (C) $-2^{2020}\omega^2$
- (D) $2^{2020}\omega^2$

2. Which of the following Argand diagrams describes the relationship defined by $\arg(z - i) = \arg(z + 1)$?



3. The acceleration of a particle moving in a straight line with velocity v is given by $\ddot{x} = \frac{1}{v}$. Which of the following functions best represents v in terms of x ?

- (A) $v = x + \frac{1}{3}$
(B) $v = \sqrt[3]{3x + 1}$
(C) $v = x + 1$
(D) $v = e^x$

4. Given $\vec{a} = \vec{i} - \vec{j} + 2\vec{k}$ and $\vec{b} = 3\vec{i} - 2\vec{j} + \vec{k}$, the projection of $\vec{a}$ onto $\vec{b}$ is:

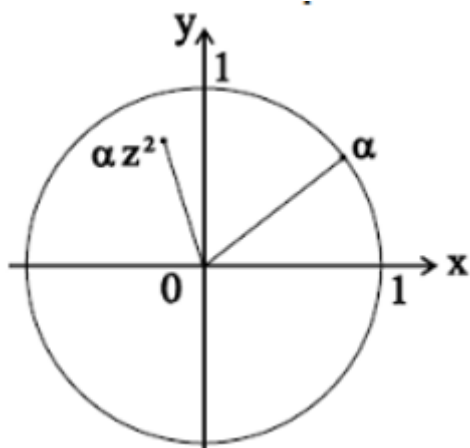
- (A) $\frac{1}{2} \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}$
(B) $\frac{7}{6} \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}$
(C) $\frac{1}{2} \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$
(D) $\frac{7}{6} \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$

5. If $a > b$ and $c > 0$, which of the following must be true?

- I. $a^2 > b^2$
II. $a - c > b - c$
III. $\frac{a}{c^2} > \frac{b}{c^2}$

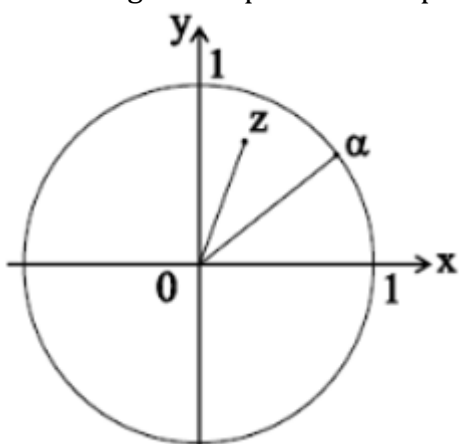
- (A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III

6. The argand diagram below shows the complex numbers α and αz^2 .

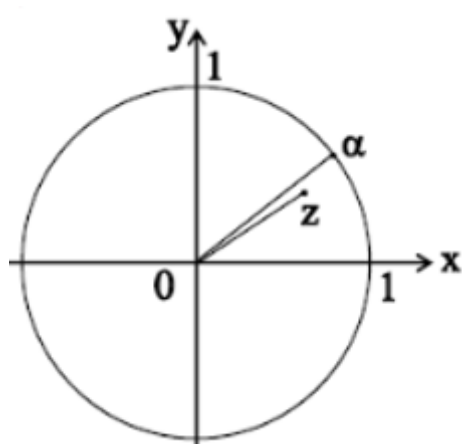


Which of the following best represents the positions of z and α ?

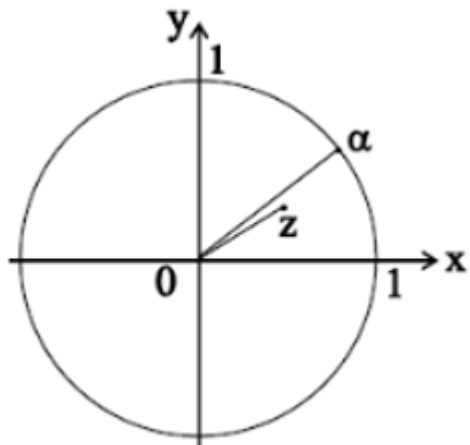
(A)



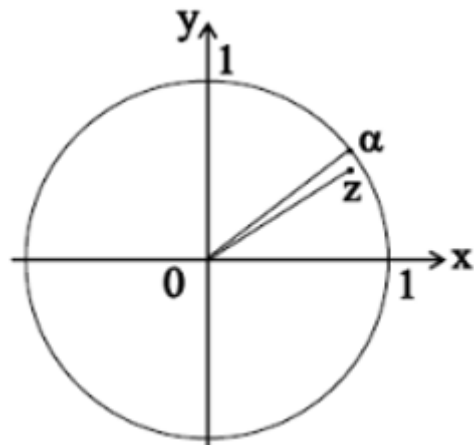
(B)



(C)



(D)



7. Which of the following expressions is equivalent to $\int_0^{2a} f(x)dx$?

(A) $\int_0^{2a} f(2a - x)dx$

(B) $\int_0^{2a} f(a - x)dx$

(C) $2 \int_0^{2a} f(2a - x)dx$

(D) $2 \int_0^{2a} f(a - x)dx$

8. What is the contrapositive of $\neg A \Rightarrow B$?

(A) $\neg B \Rightarrow \neg A$

(B) $B \Rightarrow \neg A$

(C) $\neg B \Rightarrow A$

(D) $B \Rightarrow A$

9. A sphere is defined by $\left| \underset{\sim}{r} - \begin{bmatrix} 2 \\ -3 \\ 5 \end{bmatrix} \right| = 4\sqrt{3}$.

Which of the following points lie outside of the sphere?

(A) $(2, -3, 2)$

(B) $(5, -6, 4)$

(C) $(1, 2, 3)$

(D) $(1, 0, -3)$

10. A bob on a spring moves vertically in simple harmonic motion, with equation of motion given as $x = a \sin nt$. Which of the following best describes the initial placement and motion of the bob?

- (A) the motion starts at the top and moves downwards
- (B) the motion starts at the bottom and moves upwards
- (C) the motion starts at the centre and moves upwards
- (D) the motion starts at the centre and moves downwards

End of Section I

Section II

90 marks

Attempt Questions 11–16

Allow about 2 hours and 45 minutes for this section

Answer each question in a separate writing booklet.

In Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Begin a new Writing Booklet

- (a) Find $\int \frac{dx}{x^2 + 6x + 13}$. 2
- (b) Let $w = -\sqrt{3} + i$ and $z = 1 + i$
- (i) Find wz in the form $a + ib$. 1
- (ii) Find w and z in mod-arg form. 2
- (iii) Hence, find the exact value of $\sin \frac{5\pi}{12}$. 2
- (c) Prove that the product of two consecutive even counting numbers is a multiple of 4. 2
- (d) (i) Find a, b and c such that $\frac{16}{(x^2 + 4)(2 - x)} = \frac{ax + b}{x^2 + 4} + \frac{c}{2 - x}$. 2
- (ii) Find $\int \frac{16}{(x^2 + 4)(2 - x)} dx$. 2
- (e) Solve $z^2 - 2iz + 2 = 0$. 2

End of Question 11

Question 12 (15 marks) Begin a new Writing Booklet

- (a) (i) Prove that $a^2 + b^2 \geq 2ab$. 1
- (ii) Hence, prove that $x^2 + y^2 + z^2 \geq xy + yz + zx$. 1
- (iii) Deduce that if $x + y + z = 1$, then $xy + yz + zx \leq \frac{1}{3}$. 2
- (b) Use integration by parts to show that $\int e^x \cos x \, dx = \frac{e^x}{2} (\sin x + \cos x)$. 3
- (c) Prove that $\sum_{k=1}^n r2^r = (n-1)(2^{n+1}) + 2$. 3
- (d) Sketch the region defined by the union of 3
- $$\frac{\pi}{6} \leq \arg z \leq \frac{5\pi}{6}$$
- $$|z - i| \geq 2$$
- (e) Suppose that $1 - i = e^{a+ib}$, where $a, b \in \mathbb{R}$ and $-\frac{\pi}{2} < b < \frac{\pi}{2}$. 2
Find the exact values of a and b .

End of Question 12

Question 13 (15 marks) Begin a new Writing Booklet

- (a) Find the angle between the vectors $\tilde{u} = (3, 5, -2)$ and $\tilde{v} = (-1, 2, 3)$. 2
- (b) (i) Find the square roots of $-24 - 10i$. 2
- (ii) Hence, or otherwise, solve $z^2 - (1 - i)z + 6 + 2i = 0$. 2
- (c) Consider the polynomial $P(z) = z^4 + 4z^3 + 14z^2 + 20z + 25$.
- (i) It is known that $P(-1 + 2i) = 0$. 2
Show that $P'(-1 + 2i) = 0$ and explain the significance of this.
- (ii) Explain why $-1 - 2i$ is also a root of $P(z)$. 1
- (iii) Hence, factorise $P(z)$ over the complex numbers and then over the real. 2
- (d) (i) Find the coordinates of P that divides the interval AB in the ratio 2: 3, 2
where $A = (1, 3, 4)$ and $B = (-4, 8, 2)$.
- (ii) Check the result by calculating $|\overrightarrow{AB}|$ and $|\overrightarrow{AP}|$. 2

End of Question 13

Question 14 (15 marks) Begin a new Writing Booklet

(a) Prove by contradiction that $\log_3 4$ is irrational. 2

(b) Use the substitution $u = e^x + 1$ to find $\int \frac{e^{2x}}{(e^x + 1)^2} dx$. 3

(c) (i) Show that for $k > 0$, 1

$$\frac{1}{(k+1)^2} - \frac{1}{k} + \frac{1}{k+1} < 0$$

(ii) Use mathematical induction to prove that for all integers $n \geq 2$, 3

$$\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \cdots + \frac{1}{n^2} < 2 - \frac{1}{n}.$$

(d) Using the substitution $t = \tan \frac{x}{2}$, or otherwise, find 3

$$\int \frac{dx}{5 + 3 \cos x}$$

(e) (i) Find the vector equation of $\overrightarrow{AB}$, given $A = (4, 8, 3)$ and $B = (5, 10, 4)$ 1
in the form $r_1 = \begin{bmatrix} a \\ b \\ c \end{bmatrix} + \lambda \begin{bmatrix} d \\ e \\ f \end{bmatrix}$.

(ii) Find the point of intersection of r_1 and $r_2 = \begin{bmatrix} 9 \\ 6 \\ 10 \end{bmatrix} + \mu \begin{bmatrix} 1 \\ -4 \\ 2 \end{bmatrix}$. 2

End of Question 14

Question 15 (15 marks) Begin a new Writing Booklet

- (a) Prove by the method of mathematical induction that for any positive integer $n > 0$, 3

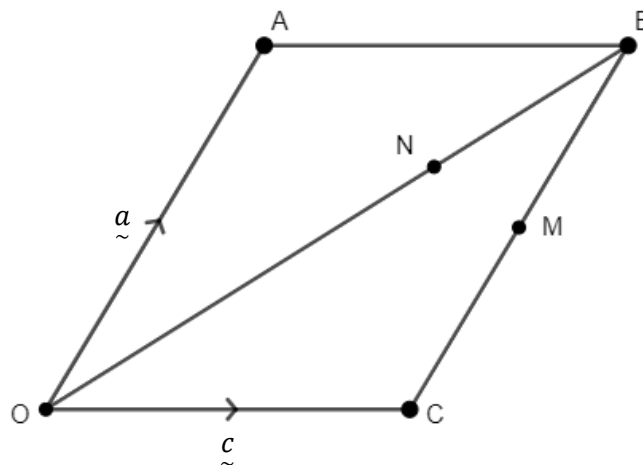
$$\frac{d}{dx}(x^n) = nx^{n-1}.$$

- (b) A particle moving in simple harmonic motion has a period of π seconds. Initially the particle is at $x = 2$ with a velocity of 8 ms^{-1} .

- (i) Find x as a function of t in the form $x = b \cos nt + c \sin nt$. 2
- (ii) Find x as a function of t in the form $x = a \cos(nt - \alpha)$, where $a > 0$ and $0 \leq \alpha \leq 2\pi$. 2
- (iii) Hence, find the amplitude and the maximum speed of the particle. 2

- (c) $OABC$ is a parallelogram. $\overrightarrow{OA}$ is represented by the vector $\underline{a}$ and $\overrightarrow{OC}$ is represented by the vector $\underline{c}$.

M is the midpoint of BC and N is the point on OB such that $ON:NB = 2:1$.



- (i) Find expressions for the following vectors, given your answers in simplest form: $\overrightarrow{ON}$, $\overrightarrow{OM}$, $\overrightarrow{AN}$ and $\overrightarrow{AM}$. 4
- (ii) Show that the points A , N and M are collinear. 2

End of Question 15

Question 16 (15 marks) Begin a new Writing Booklet

- (a) A skydiver jumps out of a stationary balloon and starts falling freely to Earth.

She experiences gravity of mg and air resistance of $\frac{mv^2}{360}$ upwards.

Given that down is *positive*, $x = t = 0$ at the balloon and that $g = 9.8 \text{ ms}^{-2}$,

- (i) Show that $\ddot{x} = g - \frac{v^2}{360}$ and find her terminal velocity. 2

- (ii) Show that $x = 180 \ln\left(\frac{360g}{360g - v^2}\right)$ and find the distance fallen when the skydiver reaches 50 ms^{-1} . 3

- (iii) Find the time taken for the skydiver to reach this speed. 3

- (b) Let $I_n = \int_0^{\frac{1}{2}} \frac{1}{(1 + 4x^2)^n} dx$, where n is a positive integer.

- (i) Find the value of I_1 . 2

- (ii) Using integration by parts, show that $I_n = \frac{2nI_{n+1}}{2n-1} + \frac{1}{2^{n+1}(1-2n)}$. 3

(Hint: $\frac{m}{(1+m)^{n+1}} = \frac{1+m-1}{(1+m)^{n+1}}$)

- (iii) Hence, evaluate $I_3 = \int_0^{\frac{1}{2}} \frac{1}{(1 + 4x^2)^3} dx$. 2

End of Examination

Q1.

$$\begin{aligned}
 \textcircled{D} &= (1 + \omega - \omega^2)^{2020} \\
 &= (1 + \omega + \omega^2 - 2\omega^2)^{2020} \\
 &= (-2\omega^2)^{2020} \\
 &= 2^{2020} \omega^{4040} \\
 &= 2^{2020} \omega^2
 \end{aligned}$$

$$\begin{array}{r}
 1346 \text{ r } 2 \\
 3 \overline{) 4040}
 \end{array}$$

Q2.

$$\textcircled{A} \quad \arg(z-i) = \arg(z+i)$$

arg of z from i same
as from $-i$

Q3

$$\begin{aligned}
 \textcircled{B} \quad \int \sqrt{x} \, dx &= \frac{2}{3} x^{3/2} + C \\
 \frac{2}{3} x^{3/2} &= x + C
 \end{aligned}$$

Q4

$$\begin{aligned}
 \textcircled{A} \quad \text{proj}_b a &= \left(\frac{a \cdot b}{b \cdot b} \right) b \\
 &= \frac{1(3) - 1(-2) + 2(1)}{3^2 + (-2)^2 + 1^2} \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix} \\
 &= \frac{7}{14} \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}
 \end{aligned}$$

Q5

(B)

$a > b \Rightarrow \sqrt{a^2} > \sqrt{b^2}$ only if $a > b > 0$
 $\Rightarrow a - c > b - c$ ✓

Q6

(D)

modulus must be closer to zero.

Q7

(A)

$$\begin{aligned} \int_0^{2a} f(2a-x) dx &= - \left[F(2a-x) \right]_0^{2a} \\ &= -F(2a-2a) + F(2a) \\ &= F(2a) - F(0) \\ \int_0^{2a} f(x) dx &= F(2a) - F(0) \end{aligned}$$

Q8

(C)

$$\begin{aligned} \neg B &\Rightarrow \neg(\neg A) \\ &\Rightarrow A \end{aligned}$$

Q9

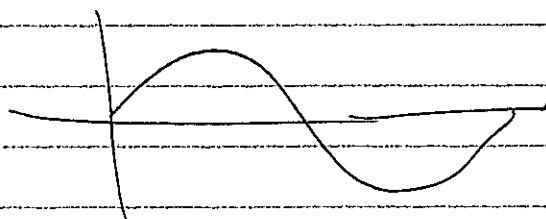
(D)

check distance from $(2, -3, 5)$
 $d^2 > 48$

$$\begin{aligned} A: d^2 &= 0^2 + 0^2 + 3^2 = 9 \\ B: d^2 &= 3^2 + (3)^2 + 1^2 = 19 \\ C: d^2 &= 1^2 + (-5)^2 + 2^2 = 30 \\ D: d^2 &= 1^2 + (-3)^2 + 8^2 = 74 \end{aligned}$$

Q10.

(C.)



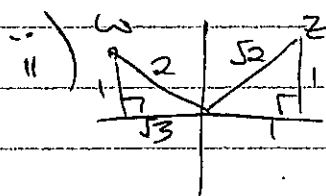
Q11 a)

$$\begin{aligned} \int \frac{dx}{x^2+6x+13} &= \int \frac{dx}{x^2+6x+9+4} \\ &= \int \frac{dx}{(x+3)^2+4} \\ &= \frac{1}{2} \tan^{-1} \frac{x+3}{2} + C \end{aligned}$$

b) i)

$$w = -\sqrt{3} + i, \quad z = 1 + i$$

$$\begin{aligned} wz &= (-\sqrt{3} + i)(1 + i) \\ &= -\sqrt{3} + i - \sqrt{3}i - 1 \\ &= -(1 + \sqrt{3}) + i(1 - \sqrt{3}) \end{aligned}$$



$$w = 2 \operatorname{cis} \frac{5\pi}{6}$$

$$z = \sqrt{2} \operatorname{cis} \frac{\pi}{4}$$

$$\text{iii) } wz = 2\sqrt{2} \operatorname{cis} \left(\frac{5\pi}{6} + \frac{\pi}{4} \right)$$

$$= 2\sqrt{2} \operatorname{cis} \left(\frac{10\pi + 3\pi}{12} \right)$$

$$= 2\sqrt{2} \operatorname{cis} \frac{13\pi}{12}$$

- ii

equating the components of i and ii

$$2\sqrt{2} \sin \frac{13\pi}{12} = (1-\sqrt{3})$$
$$\therefore \sin \frac{13\pi}{12} = \frac{1-\sqrt{3}}{2\sqrt{2}}$$

c) let two consecutive even numbers
be $2n, 2n+2$ for $n \in \mathbb{Z}$

$$\begin{aligned}\text{product} &= 2n(2n+2) \\ &= 4n^2 + 4n \\ &= 4(n^2 + n)\end{aligned}$$

hence the product is a multiple of 4

d) i) let $\frac{16}{(x^2+4)(2-x)} = \frac{ax+b}{x^2+4} + \frac{c}{2-x}$

$$\begin{aligned}\therefore 16 &= (ax+b)(2-x) + c(x^2+4) \\ &= 2ax + 2b - ax^2 - xb + cx^2 + 4c.\end{aligned}$$

for $x=2$,

$$16 = 8c$$

$$c = 2.$$

$$\begin{aligned}\text{constants: } 16 &= 2b + 4c &\Rightarrow 2b &= 8 \\ & &b &= 4\end{aligned}$$

$$\begin{aligned}\text{x-term: } 0 &= 2a - b &\rightarrow 2a &= 4 \\ & &a &= 2\end{aligned}$$

$$\therefore a=2, b=4, c=2.$$

$$\begin{aligned}
 \text{ii)} \quad \int \frac{16}{(x^2+4)(2-x)} dx &= \int \left(\frac{2x+4}{x^2+4} + \frac{2}{2-x} \right) dx \\
 &= \ln|x^2+4| + 2 \tan^{-1} \frac{x}{2} - 2 \ln|2-x| + c
 \end{aligned}$$

$$\begin{aligned}
 \text{e)} \quad z^2 - 2iz + 2 &= 0 \\
 z &= \frac{2i \pm \sqrt{(2i)^2 - 4(1)(2)}}{2(1)} \\
 &= \frac{2i \pm \sqrt{-12}}{2} \\
 &= 2i \pm 2\sqrt{3}i \\
 &= (1 \pm \sqrt{3})i, (1 - \sqrt{3})i
 \end{aligned}$$

$$\begin{aligned}
 \text{alternatively,} \\
 z^2 - 2iz - 1 &= -2 - 1 \\
 (z-i)^2 &= -3 \\
 z-i &= \pm \sqrt{3}i \\
 z &= (1 \pm \sqrt{3})i
 \end{aligned}$$

$$\begin{aligned}
 \text{Q 2 a i)} \quad &\forall a, b \in \mathbb{R} \\
 &(a-b)^2 \geq 0 \\
 &a^2 - 2ab + b^2 \geq 0 \\
 &\therefore a^2 + b^2 \geq 2ab
 \end{aligned}$$

$$\begin{aligned}
 \text{ii)} \quad &\text{from (i)} \\
 &x^2 + y^2 \geq 2xy \quad \text{--- i} \\
 &y^2 + z^2 \geq 2yz \quad \text{--- ii} \\
 &z^2 + x^2 \geq 2zx \quad \text{--- iii}
 \end{aligned}$$

i + ii + iii

$$2x^2 + 2y^2 + 2z^2 \geq 2xy + 2yz + 2zx$$

$$\therefore x^2 + y^2 + z^2 \geq xy + yz + zx$$

$$\text{iii} \quad (x+y+z)^2 = x^2 + y^2 + z^2 + 2(xy+yz+zx)$$

$$\therefore x^2 + y^2 + z^2 = 1 - 2(xy+yz+zx)$$

from (ii)

$$1 - 2(xy+yz+zx) \geq xy+yz+zx.$$

$$\therefore 3(xy+yz+zx) \leq 1$$

$$xy+yz+zx \leq \frac{1}{3}$$

$$b) \quad \int e^x \cos x dx$$

$$\text{let } u = e^x \quad du = e^x dx$$

$$dv = \cos x dx \quad v = \sin x.$$

$$= e^x \sin x - \int e^x \sin x dx$$

$$\text{let } dv = \sin x dx$$

$$v = -\cos x$$

$$= e^x \sin x - (-e^x \cos x - \int e^x (-\cos x) dx)$$

$$= e^x \sin x + e^x \cos x - \int e^x \cos x dx$$

$$\therefore 2 \int e^x \cos x dx = e^x (\sin x + \cos x)$$

$$\int e^x \cos x dx = \frac{e^x}{2} (\sin x + \cos x)$$

c) Prove. $\sum_{k=1}^n k 2^k = (n-1)(2^{n+1}) + 2.$

Prove true for $n=1$

$$\begin{aligned} \text{LHS} &= 1 \cdot 2^1 \\ &= 2 \end{aligned}$$

$$\begin{aligned} \text{RHS} &= (1-1)2^{1+1} + 2 \\ &= 2 \\ &= \text{LHS} \end{aligned}$$

$\therefore$ true for $n=1$

Assume true for $n=k$

$$\text{i.e. } \sum_{k=1}^k k 2^k = (k-1)(2^{k+1}) + 2$$

Prove true for $n=k+1$

$$\begin{aligned} \text{i.e. } \sum_{n=1}^{k+1} (k+1) 2^{k+1} &= (k+1-1)(2^{(k+1)+1}) + 2 \\ &= k 2^{k+2} + 2 \end{aligned}$$

$$\text{LHS} = (k+1)2^{k+1} + \sum_{n=1}^k n 2^n$$

$$= (k+1)2^{k+1} + (k-1)(2^{k+1}) + 2 \quad \text{by induction hypothesis}$$

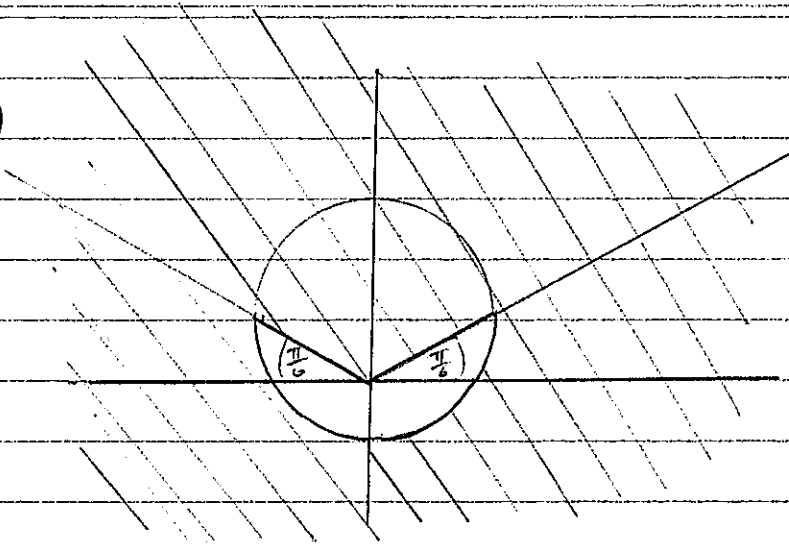
$$= (k+1 + k-1) \cdot 2^{k+1} + 2$$

$$= 2k \cdot 2^{k+1} + 2$$

$$= k 2^{k+2} + 2 = \text{RHS}$$

$\therefore$ by principle of induction, this is true

d)



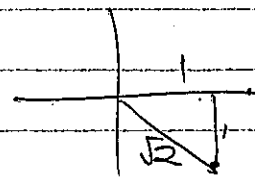
e.)

$$1-i = \sqrt{2} \operatorname{cis} \left(-\frac{\pi}{4} \right)$$

$$= \sqrt{2} e^{-\frac{\pi}{4}i}$$

$$= e^{\ln \sqrt{2}} \cdot e^{-\frac{\pi}{4}i}$$

$$= e^{\ln \sqrt{2} - \frac{\pi}{4}i}$$



$$\therefore a = \ln \sqrt{2}$$

$$b = -\frac{\pi}{4}$$

Q 13 a) $3(-1) + (5)(2) - 2(3) = \sqrt{3^2 + 5^2 + (-2)^2} \sqrt{(-1)^2 + 2^2 + 3^2} \cos \theta$

$$1 = \sqrt{38} \sqrt{14} \cos \theta$$

$$\cos \theta = \frac{1}{\sqrt{532}}$$

$$\theta = 1.527 \text{ radians}$$

$$= 87^\circ 30' 54.48''$$

b) i) let $(x+iy)^2 = -24-10i$

$$x^2 - y^2 + 2xyi = -24 - 10i$$

equate Re + Im

$$x^2 - y^2 = -24 \quad -i$$

$$2xy = -10 \Rightarrow x^2 y^2 = 25 \quad -ii$$

from i

$$x^4 - 25 = -24x^2$$

$$x^4 - 25 = -24x^2$$

$$x^4 + 24x^2 - 25 = 0$$

$$(x^2 + 25)(x^2 - 1) = 0$$

$$x^2 = 1$$

$$(x^2 \neq -25)$$

$$x = \pm 1$$

$$\Rightarrow y = \mp 5$$

$$\therefore \sqrt{-24-10i} = \pm(1 \mp 5i)$$

ii) $z^2 - (1-i)z + 6+2i = 0$

$$z = \frac{(1-i) \pm \sqrt{(1-i)^2 - 4(1)(6+2i)}}{2(1)}$$

$$= \frac{1-i \pm \sqrt{-2i - 24 - 8i}}{2}$$

$$= \frac{1-i \pm \sqrt{-24-10i}}{2}$$

$$= \frac{1-i \pm (1-5i)}{2}$$

$$= \frac{2-6i}{2}, \frac{4i}{2}$$

$$= 1-3i, 2i$$

c) $P(z) = z^4 + 4z^3 + 14z^2 + 20z + 25$

i) $P(-1+2i) = 0$

$$P'(z) = 4z^3 + 12z^2 + 28z + 20$$

$$P'(-1+2i) = 4(-1+2i)^3 + 12(-1+2i)^2 + 28(-1+2i) + 20$$

$$= 4(-1 + 3(2i) - 3(4) - 8i) + 12(1 - 4i - 4) + 28(-1+2i) + 20$$

$$= 44 - 8i + (-36) - 48i - 28 + 56i + 20$$

$$= 0$$

this means that $-1+2i$ is a double root.

ii) $-1-2i$ is also a root as

I it is the conjugate of $-1+2i$

II the polynomial has real coefficients

and complex roots of polynomials with real coefficients occur in conjugate pairs

$$\begin{aligned}
 \text{iii)} \quad \therefore P(z) &= (z - (-1-2i))^2 (z - (-1+2i))^2 \\
 &= (z+1+2i)^2 (z+1-2i)^2 \quad \text{over complex} \\
 &= ((z+1)^2 - (2i)^2)^2 \\
 &= (z^2 + 2z + 1 - (-4))^2 \\
 &= (z^2 + 2z + 5)^2 \quad \text{over real}
 \end{aligned}$$

$$\begin{aligned}
 \text{d) i)} \quad A &= (1, 3, 4) & AB &\Rightarrow 2:3 \\
 B &= (-4, 8, 2)
 \end{aligned}$$

$$\begin{aligned}
 P &= \frac{1}{2+3} (3(1, 3, 4) + 2(-4, 8, 2)) \\
 &= \frac{1}{5} ((3, 9, 12) + (-8, 16, 4)) \\
 &= \frac{1}{5} (-5, 25, 16) \\
 &= (-1, 5, 3\frac{1}{5})
 \end{aligned}$$

$$\begin{aligned}
 \text{ii)} \quad \vec{AB} &= \begin{bmatrix} -4 \\ 8 \\ 2 \end{bmatrix} - \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix} & \vec{AP} &= \begin{bmatrix} -1 \\ 5 \\ \frac{16}{5} \end{bmatrix} - \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix} \\
 &= \begin{bmatrix} -5 \\ 5 \\ -2 \end{bmatrix} & &= \begin{bmatrix} -2 \\ 2 \\ -\frac{4}{5} \end{bmatrix}
 \end{aligned}$$

$$\begin{aligned}
 |\vec{AB}| &= \sqrt{(-5)^2 + 5^2 + (-2)^2} \\
 &= \sqrt{54}
 \end{aligned}$$

$$\begin{aligned}
 |\vec{AP}| &= \sqrt{(-2)^2 + 2^2 + (-\frac{4}{5})^2} \\
 &= \sqrt{8 + \frac{16}{25}}
 \end{aligned}$$

$$\begin{aligned}
 |\vec{AP}| &= \sqrt{\frac{216}{25}} \\
 &= \frac{2}{5} \sqrt{54} \\
 &= \frac{2}{5} |\vec{AB}|
 \end{aligned}$$

Q14.

a) let $\log_3 4 = \frac{a}{b}$, $a, b \in \mathbb{Z}$

$$b \log_3 4 = a$$

$$\log_3 4^b = a$$

$$4^b = 3^a$$

$$2^{2b} = 3^a$$

given the primeness of 2 and 3, and $a, b \in \mathbb{Z}$,
 this cannot be true.
 $\therefore \log_3 4$ is irrational

b) $I = \int \frac{e^{2x}}{(e^x + 1)^2} dx$ let $u = e^x + 1$, $e^x = u - 1$,
 $du = e^x dx$

$$\begin{aligned}
 &= \int \frac{e^x}{(e^x + 1)^2} \cdot e^x dx \\
 &= \int \frac{u-1}{u^2} du \\
 &= \int \left(\frac{1}{u} - \frac{1}{u^2} \right) du
 \end{aligned}$$

$$= \ln|a| + \frac{1}{a} + C$$

$$= \ln(e^x + 1) + \frac{1}{e^x + 1} + C$$

c) i. Show $\frac{1}{(k+1)^2} - \frac{1}{k} + \frac{1}{k+1} < 0$

$$\text{LHS} = \frac{1}{(k+1)^2} - \frac{1}{k} + \frac{1}{k+1}$$

$$= \frac{k - (k+1)^2 + k(k+1)}{k(k+1)^2}$$

$$= \frac{k - k^2 - 2k - 1 + k^2 + k}{k(k+1)^2}$$

$$= \frac{-1}{k(k+1)^2} < 0 \text{ as } k > 0$$

ii) Prove $\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{n^2} < 2 - \frac{1}{n}, \quad n \geq 2$

Prove true for $n=2$

$$\text{LHS} = \frac{1}{1^2} + \frac{1}{2^2}$$

$$= \frac{5}{4}$$

$$\text{RHS} = 2 - \frac{1}{n}$$

$$= \frac{3}{2}$$

as $\frac{5}{4} < \frac{3}{2}$, then statement true for $n=2$

Assume true for $n=k$

$$\therefore \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{k^2} < 2 - \frac{1}{k} \quad \text{— induction hypothesis}$$

Prove true for $n=k+1$

$$\text{F.R. } \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{k^2} + \frac{1}{(k+1)^2} < 2 - \frac{1}{k+1}$$

$$\text{LHS} < 2 - \frac{1}{k} + \frac{1}{(k+1)^2}, \text{ using induction hypothesis}$$

$$= 2 + \frac{1}{(k+1)^2} - \frac{1}{k} + \frac{1}{k+1} - \frac{1}{k+1}$$

$$< 2 - \frac{1}{k+1} \quad \text{from (i)}$$

$$< \text{RHS}$$

$\therefore$ by principle of induction, statement is true for $n \geq 2$

$$\text{Q) } I = \int \frac{dx}{5 + 3 \cos x} \quad \text{let } t = \tan \frac{x}{2} \quad + \frac{\sqrt{1+t^2}}{t \cdot \frac{x}{2}}$$

$$\therefore \cos x = \frac{1-t^2}{1+t^2} \quad \frac{dt}{dx} = \frac{1}{2} \sec^2 \frac{x}{2}$$

$$dx = \frac{2dt}{1+t^2}$$

$$I = \int \frac{1}{5 + \frac{3(1-t^2)}{1+t^2}} \cdot \frac{2dt}{1+t^2}$$

$$= 2 \int \frac{\cancel{1+t^2} dt}{(5(1+t^2) + 3(1-t^2)) \cancel{(1+t^2)}}$$

$$= 2 \int \frac{dt}{8 + 2t^2}$$

$$= \int \frac{dt}{4 + t^2}$$

$$= \frac{1}{2} \tan^{-1} \frac{1}{2} + C.$$

$$= \frac{1}{2} \tan^{-1} \left[\frac{\tan \frac{x}{2}}{2} \right] + C$$

c) i) $A = (4, 8, 3), B = (5, 10, 4)$

$$\underline{r}_1 = A + \lambda \overrightarrow{AB}$$

$$= \begin{bmatrix} 4 \\ 8 \\ 3 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

ii) equate $\underline{r}_1$ and $\underline{r}_2$

$$\begin{bmatrix} 4 \\ 8 \\ 3 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 9 \\ 6 \\ 10 \end{bmatrix} + \mu \begin{bmatrix} 1 \\ -4 \\ 2 \end{bmatrix}$$

$$\therefore 4 + \lambda = 9 + \mu \quad \text{--- i'}$$

$$8 + 2\lambda = 6 - 4\mu \quad \text{--- ii'}$$

$$3 + \lambda = 10 + 2\mu \quad \text{--- iii'}$$

i - iii

$$1 = -1 - \mu$$

$$\mu = -2 \Rightarrow \lambda = 3 \quad (\text{from i})$$

test in ii LHS = $8 + 2(3)$ RHS = $6 - 4(-2)$
 $= 14$ $= 14$

iii LHS = $3 + 3$ RHS = $10 + 2(-2)$
 $= 6$ $= 6$

$$\therefore \text{intersection at } \begin{bmatrix} 4 \\ 8 \\ 3 \end{bmatrix} + 3 \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 7 \\ 14 \\ 6 \end{bmatrix}$$

Q15.a) Prove $\frac{d}{dx}(x^n) = nx^{n-1}$.

Prove true for $n=$

$$\begin{aligned}\text{LHS} &= \frac{d}{dx}(x^1) \\ &= 1\end{aligned}$$

$$\begin{aligned}\text{RHS} &= 1 \cdot x^0 \\ &= 1 \\ &= \text{LHS}\end{aligned}$$

$\therefore$ true for $n=1$

Assume true for $n=k$

$$\therefore \frac{d}{dx}(x^k) = kx^{k-1}$$

Prove true for $n=k+1$

$$\text{i.e. } \frac{d}{dx}(x^{k+1}) = (k+1)x^{(k+1)-1}$$

$$\text{LHS} = \frac{d}{dx}(x^{k+1})$$

$$= \frac{d}{dx}(x \cdot x^k)$$

$$= x^k \cdot \frac{d}{dx}(x) + x \cdot \frac{d}{dx}(x^k)$$

$$= x^k \cdot 1 + x \cdot kx^{k-1} \quad \text{using induction hypothesis}$$

$$= x^k + kx^k$$

$$= (k+1)x^k$$

$$= (k+1)x^{(k+1)-1} = \text{RHS}$$

$\therefore$ by induction, true for $n \geq 1$

$$b) \quad i) \quad x = b \cos nt + c \sin nt$$

$$t=0, x=2$$

$$2 = b \cos 0 + c \sin 0 \\ = b.$$

$$v = -bn \sin nt + cn \cos nt$$

$$t=0, v=8$$

$$8 = -bn \sin 0 + cn \cos 0$$

$$\therefore 8 = cn$$

as period is π , $n=2$

$$\therefore b=2, c=4$$

$$\text{and } x = 2 \cos 2t + 4 \sin 2t$$

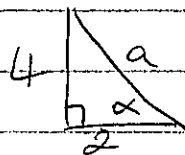
$$ii) \quad x = a \cos(nt - \alpha)$$

$$= a \cos nt \cdot \cos \alpha + a \sin nt \cdot \sin \alpha.$$

equating coefficients

$$a \cos \alpha = 2$$

$$a \sin \alpha = 4$$



$$\therefore a = 2\sqrt{5}$$

$$\tan \alpha = 2 \Rightarrow \alpha = \tan^{-1} 2, \quad 0 < \alpha < \frac{\pi}{2}$$

as $\cos \alpha > 0$
 $\sin \alpha > 0$

$$\therefore x = 2\sqrt{5} \cos(2t - \tan^{-1} 2)$$

iii. for SHM, max speed when $x=0$.

$$\Rightarrow 2\sqrt{5} \cos(2t - \tan^{-1} 2) = 0$$

$$2t - \tan^{-1} 2 = 1$$

$$t = \frac{1}{2}(1 + \tan^{-1} 2)$$

$$v = -4\sqrt{5} \sin(2t - \tan^{-1} 2)$$

$$= -4\sqrt{5} \sin\left(2 \cdot \frac{1}{2}(1 + \tan^{-1} 2) - \tan^{-1} 2\right)$$

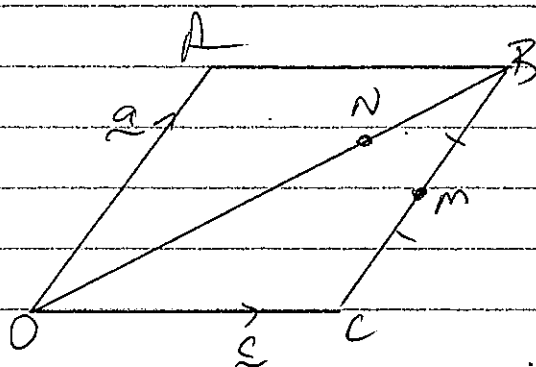
$$= -4\sqrt{5} \sin 1$$

$$= -7.5263453$$

$\therefore$ amplitude is $2\sqrt{5}$ units

and max speed is 7.53 ms^{-1} (2dp)

Q i.



$$\vec{ON} = \frac{2}{3}(\vec{a} + \vec{c})$$

$$\vec{OM} = \frac{1}{2}\vec{a} + \vec{c}$$

$$\begin{aligned} \vec{AN} &= \vec{ON} - \vec{OA} \\ &= \frac{2}{3}(\vec{a} + \vec{c}) - \vec{a} \end{aligned}$$

$$= -\frac{1}{3}\vec{a} + \frac{2}{3}\vec{c}$$

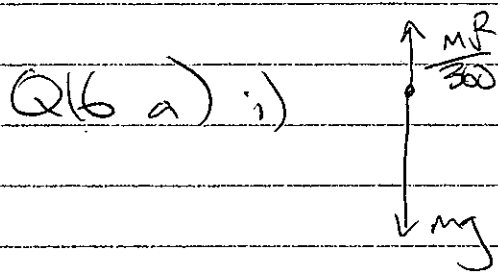
$$\begin{aligned} \vec{AM} &= \vec{OM} - \vec{OA} \\ &= \frac{1}{2}\vec{a} + \vec{c} - \vec{a} \end{aligned}$$

$$= -\frac{1}{2}\vec{a} + \vec{c}$$

$$\begin{aligned}
 \text{ii)} \quad \vec{AN} &= -\frac{1}{3}\underline{a} + \frac{2}{3}\underline{c} \\
 &= \frac{2}{3}\left(-\frac{1}{2}\underline{a} + \underline{c}\right) \\
 &= \frac{2}{3}\vec{AM}
 \end{aligned}$$

as both vectors originate at A.

$\therefore$ A, N, and M are collinear.



$$F = ma$$

$$ma = mg - \frac{mv^2}{360}$$

$$a = g - \frac{v^2}{360}$$

for terminal velocity, $a = 0$

$$\therefore g = \frac{v^2}{360}$$

$$v^2 = 360g$$

$$v = \sqrt{360g}$$

$$= 59.39696 \text{ ms}^{-1}$$

$$\begin{aligned}
 \text{ii)} \quad \frac{v dv}{dx} &= g - \frac{v^2}{360} \\
 &= \frac{360g - v^2}{360}
 \end{aligned}$$

$$\int \frac{v dv}{360g - v^2} = \int \frac{dx}{360}$$

$$-\frac{1}{2} \ln |360g - v^2| = \frac{x}{360} + c$$

given $t=0 \Rightarrow x=0, v=0$

$$\text{then } c = -\frac{1}{2} \ln 360g$$

$$\frac{x}{360} = \frac{1}{2} \ln 360g - \frac{1}{2} \ln (360g - v^2)$$

$$= \frac{1}{2} \ln \frac{360g}{360g - v^2}$$

$$\therefore x = 180 \ln \frac{360g}{360g - v^2}$$

at $v=50$

$$x = 180 \ln \frac{360g}{360g - 50^2}$$

$$= 221.9608748 \text{ m below the balloon}$$

ii) $\ddot{x} = \frac{360g - v^2}{360}$

$$\therefore \frac{dv}{dt} = \frac{360g - v^2}{360}$$

$$dt = \frac{360 dv}{360g - v^2}$$

$$t = \int \frac{360 dv}{(360g - v^2)}$$

$$\text{let } \frac{360}{360g - v^2} = \frac{A}{\sqrt{360g - v^2}} + \frac{B}{\sqrt{360g + v^2}}$$

$$\therefore 360 = A(\sqrt{360g + v^2}) + B(\sqrt{360g - v^2})$$

$$\text{when } v = \sqrt{360g} \quad A = \frac{360}{2\sqrt{360g}} \\ = \frac{3\sqrt{10}}{\sqrt{g}}$$

$$\text{when } v = -\sqrt{360g} \quad B = \frac{3\sqrt{10}}{\sqrt{g}}$$

$$\therefore t = \int \left(\frac{3\sqrt{10}}{\sqrt{g}(\sqrt{360g} - v)} + \frac{3\sqrt{10}}{\sqrt{g}(\sqrt{360g} + v)} \right) dv \\ = \frac{3\sqrt{10}}{\sqrt{g}} \left(-\ln(\sqrt{360g} - v) + \ln(\sqrt{360g} + v) \right) + c$$

$$= \frac{3\sqrt{10}}{\sqrt{g}} \ln \frac{\sqrt{360g} + v}{\sqrt{360g} - v} + c$$

$$\text{at } t=0, v=0 \Rightarrow c=0 \quad (\ln 1=0)$$

$$\text{at } v=50$$

$$t = \frac{3\sqrt{10}}{\sqrt{g}} \ln \frac{\sqrt{360g} + 50}{\sqrt{360g} - 50}$$

$$= 7.438548783$$

$\therefore$ skydiver reaches 50 ms^{-1} after 7.44 s (2dp)

$$(b) \quad i. \quad I_n = \int_0^{\frac{1}{2}} \frac{1}{(1+4x^2)^n} dx$$

$$I_1 = \int_0^{\frac{1}{2}} \frac{1}{1+4x^2} dx$$

$$= \frac{1}{2} \left[\tan^{-1} 2x \right]_0^{\frac{1}{2}}$$

$$= \frac{1}{2} \left(\tan^{-1} 2\left(\frac{1}{2}\right) - \tan^{-1} 2(0) \right)$$

$$= \frac{1}{2} \left(\frac{\pi}{4} - 0 \right)$$

$$= \frac{\pi}{8}$$

$$ii. \quad \text{let } u = (1+4x^2)^{-n} \quad dv = dx$$

$$du = -n(8x)(1+4x^2)^{-n-1} dx \quad v = x.$$

$$= \frac{-8nx}{(1+4x^2)^{n+1}}$$

$$I_n = \left[\frac{x}{(1+4x^2)^n} \right]_0^{\frac{1}{2}} + 2n \int_0^{\frac{1}{2}} \frac{4x^2}{(1+4x^2)^{n+1}} dx.$$

$$= \frac{\frac{1}{2}}{(1+4(\frac{1}{2})^2)^n} - \frac{0}{(1+4(0)^2)^n} + 2n \int_0^{\frac{1}{2}} \frac{1+4x^2-1}{(1+4x^2)^{n+1}} dx$$

$$= \frac{1}{2 \times 2^n} + 2n \int_0^{\frac{1}{2}} \frac{1+4x^2}{(1+4x^2)^{n+1}} dx - 2n \int_0^{\frac{1}{2}} \frac{1}{(1+4x^2)^{n+1}} dx$$

$$= \frac{1}{2^{n+1}} + 2^n \int_0^{\frac{1}{2}} \frac{dx}{(1+4x^2)^n} - 2^n \int_0^{\frac{1}{2}} \frac{dx}{(1+4x^2)^{n+1}}$$

$$\therefore I_n = \frac{1}{2^{n+1}} + 2^n I_n - 2^n I_{n+1}$$

$$(2n-1)I_n = 2^n I_{n+1} - \frac{1}{2^{n+1}}$$

$$\therefore I_n = \frac{2^n I_{n+1}}{2n-1} - \frac{1}{2^{n+1}(2n-1)}$$

$$= \frac{2^n I_{n+1}}{2n-1} + \frac{1}{2^{n+1}(1-2n)}$$

$$\text{iii) } I_1 = \frac{2(1) I_2}{2(1)-1} + \frac{1}{2^{1+1}(1-2(1))}$$

$$\frac{\pi}{8} = 2I_2 - \frac{1}{4}$$

$$I_2 = \frac{1}{2} \left(\frac{\pi}{8} + \frac{1}{4} \right)$$

$$= \frac{\pi+2}{16}$$

$$I_2 = \frac{2(2) I_3}{2(2)-1} + \frac{1}{2^{2+1}(1-2(2))}$$

$$\frac{\pi+2}{16} = \frac{4I_3}{3} - \frac{1}{3 \times 2^3}$$

$$\begin{aligned} \frac{4}{3} I_3 &= \frac{\pi+2}{16} + \frac{1}{24} \\ &= \frac{3\pi+6+2}{48} \end{aligned}$$

$$I_3 = \frac{\cancel{3} (3\pi+8)}{4 \cancel{48} 16}$$

$$= \frac{3\pi+8}{64}$$